Analytical Chemistry Cumulative Exam
Feb. 29, 2008
(rounding, significant figures, and basic electrochemistry)
Note: constants and half reactions are provided at the end of this cume.

1. (12 points) Round the result of the following equation to the appropriate number of significant figures.

$$y = (0.034 - 4.0 \times 10^{-3})/(22.8 + 0.2)$$

2. (18 points) Round the result of the following calculation, paying careful attention to the ***exact*** number of significant figures. (Use the meticulous "fraction" approach to get the number of significant figures right in the answer here).

$$y = 0.1234 / (143.321*0.800)$$

3. (10 points) What is the pH of a solution that has an activity of hydrogen, H^+, (or hydronium, H_3O^+) ions of 2.34×10^{-7} M? (Pay attention to significant figures!)

4. (20 points) What is the overall cell reaction and the emf in the following system? Is the reaction spontaneous? Is it galvanic or electrolytic? Assume that the system is aqueous.

$$Ag \mid AgCl \mid K^+, Cl^- (1\ M) \parallel K^+, Cl^- (0.1\ M) \mid AgCl \mid Ag$$

5. (13 points) For a given redox reaction between H_2SO_3 and MnO_4^{2-}, what is the overall balanced chemical equation? Put appropriate coefficients in the blanks and add H^+ and H_2O as needed. If a coefficient is "1", please write "1" in the blank. (You will have to transcribe this problem into your blue book and put the answers there).

$$____\ H_2SO_3 + ____\ MnO_4^{2-} \rightarrow ____\ SO_4^{2-} + ____\ Mn^{2+} + ___\ _____$$

These questions will help you balance the equation above:

What is the oxidation state of the S in H_2SO_3? _____

What is the oxidation state of the Mn in MnO_4^{2-}? _____

How many electrons are involved in converting H_2SO_3 to SO_4^{2-}? _____

How many electrons are involved in converting MnO_4^{2-} to Mn^{2+}? _____

Which of the reactant species starts out in the reduced form? _____

Which reactant species is the reductant? _____

6. (17 points) Calculate the equilibrium constant, K_{redox}, for the following reaction. (Show how you derive the expression of K_{redox} as a function of the redox potentials.) (Assume that the [H+] = 1.50 x 10^{-3} M, if you need it.)

$$VO^{2+} + V^{2+} + 2H^{+} = 2V^{3+} + H_2O$$

Use the following half reactions:

$V^{3+} + e- = V^{2+}$	E° = -0.255 V vs. NHE
$VO^{2+} + 2H^{+} + e^{-} = V^{3+} + H_2O$	E° = +0.359 V vs. NHE

7. (10 points) The E° value of the following reaction in pH 9 solution is about 0.000 V vs. Ag/AgCl (saturated KCl).

$$p\text{-quinoneimine } (PAP_O) + 2\ e^{-} + 2\ H^{+} = p\text{-aminophenol } (PAP_R)$$

What is the E° value for this half reaction vs. NHE?

Tables and Constants for Analytical Cume

The following might or might not be helpful to you in solving the problems:

RT/F = 0.02569 V at 25 °C
F/RT = 38.92 V^{-1} at 25 °C
F = 96485.3 Coul/mole e^-
R = 8.31447 J mol^{-1} K^{-1}

$Ag^+ + e = Ag$	E° = 0.7991 V vs. NHE
$AgBr + e = Ag + Br^-$	E° = 0.0711 V vs. NHE
$AgCl + e = Ag + Cl^-$	E° = 0.2223 V vs. NHE
$AgCl + e = Ag + Cl^-$ (saturated KCl)	E° = 0.197 V vs. NHE
$AgI + e = Ag + I^-$	E° = -0.1522 V vs. NHE
$Ag_2O + H_2O + 2e = 2\ Ag + 2\ OH^-$	E° = 0.342 vs. NHE
$Cl_{2(g)} + 2e = 2\ Cl^-$	E° = 1.3583 V vs. NHE
$2\ H^+ + 2\ e = H_2$	E° = 0.0000 V vs. NHE
$Hg_2Cl_2 + 2\ e = 2\ Hg + 2\ Cl^-$ (saturated KCl)	E° = 0.2415 vs. NHE
$K^+ + e = K$	E° = -2.925 V vs. NHE